Speech Command Recognition + RL Threshold Tuning

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1. PROBLEM STATEMENT & SCOPE

Speech-based interfaces are increasingly used in everyday systems such as virtual assistants, smart devices, and accessibility tools. However, one persistent challenge is ensuring reliable recognition of short spoken commands under varying conditions (e.g., noise, accents, microphone quality). Misclassifications can lead to incorrect system behavior, while overly strict filtering may reject valid inputs.

This project aims to develop a speech command recognition system that classifies short spoken commands such as "yes," "no," "stop," and "go" using a Convolutional Neural Network (CNN). On top of this, a Reinforcement Learning (RL) agent will be introduced to dynamically determine the optimal confidence threshold for accepting or rejecting predictions.

The key idea is that instead of using a fixed confidence threshold, the RL agent learns a policy that balances the trade-off between:

1. False Accepts (incorrect predictions accepted as correct)
2. False Rejects (correct predictions rejected as uncertain)

Scope of the Project:

1. Focus on a subset of ~10 command words from a larger dataset.
2. Train a CNN model on audio spectrograms.
3. Implement an RL-based threshold tuning mechanism.
4. Evaluate performance using classification metrics and decision cost.

This project does not aim to build a full real-time voice assistant but instead focuses on improving the decision reliability layer of speech classification systems.

2. RELATED WORK

Speech command recognition has been widely studied, particularly with the release of datasets such as the Google Speech Commands Dataset, which enabled standardized benchmarking of keyword spotting systems.

CNN-based approaches have shown strong performance in audio classification tasks by converting audio signals into spectrograms. Architectures such as ResNet have been adapted for speech recognition due to their ability to capture hierarchical features and improve training stability.

In recent years, research has explored combining classification models with decision-making frameworks. Reinforcement Learning, particularly approaches like Q-learning and contextual bandits, has been used to optimize thresholds, resource allocation, and decision policies in uncertain environments.

Prior work suggests that static thresholds are suboptimal in dynamic or noisy settings. Adaptive thresholding using RL provides a promising solution by learning from feedback signals and minimizing long-term cost rather than optimizing only immediate accuracy.

This project builds on these ideas by integrating CNN-based speech classification with RL-driven decision optimization.

3. DATASET PLAN

The primary dataset for this project is the Google Speech Commands Dataset, which contains thousands of one-second audio clips of spoken words from a diverse group of speakers.

Dataset Characteristics:

a. 35 different command words
b. Thousands of labeled audio samples
c. Variability in speaker demographics and recording conditions

Planned Usage:

a. Select a subset of approximately 10 commands (e.g., "yes," "no," "stop," "go," etc.)
b. Convert raw audio signals into Mel spectrograms for model input
c. Split data into training, validation, and testing sets

Preprocessing Steps:

a. Normalize audio signals
b. Generate spectrograms using Short-Time Fourier Transform (STFT)
c. Apply augmentation (optional): noise injection, time shifting

This dataset is chosen due to its accessibility, standard usage in research, and suitability for lightweight speech recognition tasks.

4. REINFORCEMENT LEARNING COMPONENTS

4.1 CNN Model (Speech Classifier)

A Convolutional Neural Network will be used to classify speech commands from spectrogram inputs.

Design Options:

a. A lightweight custom 3-layer CNN, or
b. A modified ResNet architecture

Input: Spectrogram images

Output: Probability distribution over command classes

Evaluation Metrics:

1. Accuracy
2. Macro F1-score

4.2 NLP Component (Command Metadata Analysis)

Although the input is audio, a lightweight NLP component will analyze command labels and metadata.

Purpose:

a. Group commands by semantic similarity (e.g., action vs. confirmation)
b. Provide contextual signals to the RL agent

This component may use simple techniques such as label encoding or embeddings to represent command relationships.

4.3 RL Component (Threshold-Tuning Agent)

The RL agent will learn to select an optimal confidence threshold for accepting or rejecting CNN predictions.

Approach:

1. Use Q-learning or a contextual bandit approach
2. State: Confidence score and/or contextual features
3. Action: Adjust threshold (increase, decrease, or maintain)
4. Reward: Based on classification outcome and associated cost

Reward Design:

1. Penalize false accepts more heavily (if safety-critical)
2. Penalize false rejects moderately
3. Reward correct acceptances

Evaluation Metric:

1. Expected cost (combination of false accept and false reject penalties)

This component enables adaptive decision-making, improving robustness compared to fixed-threshold systems.

5. ETHICS RISK REGISTER

1. **Accent/dialect bias** — model may perform worse on non-native English speakers. Mitigation: report per-accent accuracy if data allows.
2. **Privacy** — audio recording without consent. Mitigation: only use public datasets, add consent disclaimers.
3. **Accessibility** — may fail for people with speech impairments. Mitigation: add confidence score display and fallback options.

6. REFERENCES

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